

Chi-Square Calculation: An Example

	Play chess	Not play chess	Sum (row)
Like science fiction	250 (X1)	200 (X2)	450
Not like science fiction	50 (X3)	1000 (X4)	1050
Sum(col.)	300	1200	1500

- ❑ Null hypothesis: The two distributions are independent
 - ❑ What does that mean?
 - ❑ The ratio between people who play chess vs not play chess is the same for both groups of like science fiction and not like science fiction
 - ❑ $X1:X2=X3:X4=300:1200$
 - ❑ $X1:X3=X2:X4=450:1050$
 - ❑ $X1+X2=450$ $X3+X4=1050$
 - ❑ $X1+X3=300$ $X2+X4=1200$

Chi-Square Calculation: An Example

	Play chess	Not play chess	Sum (row)
Like science fiction	250 (90)	200 (360)	450
Not like science fiction	50 (210)	1000 (840)	1050
Sum(col.)	300	1200	1500

How to derive 90?
 $450/1500 * 300 = 90$

- χ^2 (chi-square) calculation (numbers in parenthesis are expected counts calculated based on the data distribution in the two categories)

$$\chi^2 = \frac{(250 - 90)^2}{90} + \frac{(50 - 210)^2}{210} + \frac{(200 - 360)^2}{360} + \frac{(1000 - 840)^2}{840} = 507.93$$


We can reject the null hypothesis of independence at a confidence level of 0.001

- It shows that like_science_fiction and play_chess are correlated in the group

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	A	B	C	D	Sum (row)
1					200
0					1000
Sum(col.)	300	300	300	300	1200

□ Degree of freedom

□ $(\text{\#categories_in_variable_A} - 1) \times (\text{\#categories_in_variable_B} - 1)$

□ number of values that are free to vary

Chi-Square Calculation: An Example

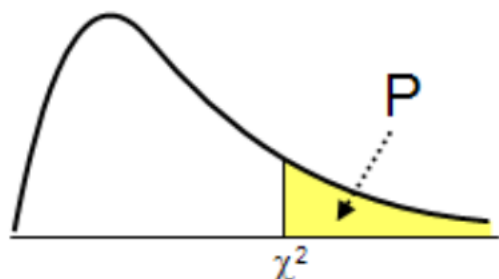
	Play chess	Not play chess	Sum (row)
Like science fiction	250 (90)	200 (360)	450
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$$\chi^2 = \frac{(250 - 90)^2}{90} + \frac{(50 - 210)^2}{210} + \frac{(200 - 360)^2}{360} + \frac{(1000 - 840)^2}{840} = 507.93$$

□ Degree of freedom =?

We can reject the null hypothesis of independence at a confidence level of 0.001

Values of the Chi-squared distribution



	P										
DF	0.995	0.975	0.20	0.10	0.05	0.025	0.02	0.01	0.005	0.002	0.001
1	0.0000393	0.000982	1.642	2.706	3.841	5.024	5.412	6.635	7.879	9.550	10.828
2	0.0100	0.0506	3.219	4.605	5.991	7.378	7.824	9.210	10.597	12.429	13.816
3	0.0717	0.216	4.642	6.251	7.815	9.348	9.837	11.345	12.838	14.796	16.266
4	0.207	0.484	5.989	7.779	9.488	11.143	11.668	13.277	14.860	16.924	18.467
5	0.412	0.831	7.289	9.236	11.070	12.833	13.388	15.086	16.750	18.907	20.515
6	0.676	1.237	8.558	10.645	12.592	14.449	15.033	16.812	18.548	20.791	22.458